

**CENTRAL BOARD OF SECONDARY EDUCATION****HALF-YEARLY PRACTICE EXAMINATION (SESSION 2024-2025)****SUBJECT: SCIENCE (CLASS - IX) | CHAPTER 5: EXPLORING MIXTURES AND THEIR SEPARATION [SET B - ADVANCED HOTS]****TIME ALLOWED: 45 MINUTES****MAXIMUM MARKS: 25****PAPER CODE: 086/CH05/SET-B****Student's Name:** \_\_\_\_\_**Roll No.:** \_\_\_\_\_**Class & Section:** Class IX - \_\_\_\_\_**Date of Exam:** \_\_\_\_\_**Invigilator's Sign:** \_\_\_\_\_**GENERAL INSTRUCTIONS (SET B - HIGHER ORDER THINKING SKILLS):**

1. The question paper comprises **25 Multiple Choice Questions (MCQs)** of **1 mark each**.
2. **Section A (Q1 – Q15):** Advanced Conceptual and Numerical MCQs with single correct option.
3. **Section B (Q16 – Q20):** Critical Assertion-Reasoning based questions testing analytical clarity.
4. **Section C (Q21 – Q25):** Case-Study / Practical Experimental Data-interpretation questions.
5. All questions are compulsory. There is no negative marking.
6. Darken the corresponding circle on the **OMR Sheet** completely using a blue/black ballpoint pen.

**SECTION A: ADVANCED & NUMERICAL MCQS (Q.1 TO Q.15)****[15 MARKS]**

**Q.1** The solubility of potassium nitrate in water at 60°C is 106 g per 100 g of water, and at 20°C it is 32 g per 100 g of water. If a saturated solution prepared with 50 g of water at 60°C is cooled down to 20°C, what mass of potassium nitrate crystals will precipitate out?

- (A) 74 g (B) 37 g  
(C) 53 g (D) 16 g

**Q.2** How much common salt must be added to 180 g of water to prepare an exact 10% (mass by mass) saline solution?

- (A) 18 g (B) 20 g  
(C) 10 g (D) 25 g

**Q.3** In paper chromatography, a dye spot travels 4.8 cm from the baseline while the solvent front travels 8.0 cm. What is the Retention Factor ( $R_f$ ) of the dye?

- (A) 1.67 (B) 0.60  
(C) 0.48 (D) 0.80

**Q.4** Which sequence of methods should be employed to separate a solid mixture containing Ammonium Chloride, Sand, and Common Salt into pure individual components?

- (A) Filtration → Sublimation → Distillation (B) Sublimation (recovers  $\text{NH}_4\text{Cl}$ ) → Dissolution in water & Filtration (separates sand) → Evaporation/Crystallization (recovers salt)  
(C) Magnetic separation → Chromatography → Decantation (D) Centrifugation → Sublimation → Fractional distillation

**Q.5** What is the primary role of glass beads packed inside a fractionating column during fractional distillation of miscible liquids?

- (A) To react chemically with lower boiling components (B) To provide an extensive surface area for repeated cycles of condensation and re-vaporization  
(C) To absorb water vapor selectively (D) To increase atmospheric pressure inside the column

**Q.6 Which of the following colloidal systems represents a 'gel'?**

- |                               |                           |
|-------------------------------|---------------------------|
| (A) Fog and mist              | (B) Milk and face cream   |
| (C) Jelly, butter, and cheese | (D) Pumice stone and foam |

**Q.7 Colloidal particles in a sol do not settle down under the pull of gravity over weeks primarily because of:**

- |                                         |                                                                                                                              |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| (A) Gravitational repulsion             | (B) Continuous random zig-zag bombardment by solvent molecules (Brownian motion) and identical electric charges on particles |
| (C) Extremely high density of particles | (D) Zero kinetic energy of particles                                                                                         |

**Q.8 Which of the following processes is a chemical change rather than a physical change?**

- |                                   |                                                   |
|-----------------------------------|---------------------------------------------------|
| (A) Dissolving sugar in hot water | (B) Rusting of an iron bicycle frame in moist air |
| (C) Melting of ice cubes          | (D) Boiling of liquid nitrogen                    |

**Q.9 Why is crystallization considered a superior purification technique compared to evaporation to dryness for copper sulphate?**

- |                                               |                                                                                                             |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| (A) Crystallization consumes more electricity | (B) In evaporation, solids may decompose or char and soluble impurities remain mixed with the dried residue |
| (C) Crystallization requires toxic solvents   | (D) Crystallization destroys crystal structures                                                             |

**Q.10 A mixture of acetone (boiling point 56°C) and water (boiling point 100°C) is separated by simple distillation. Which component distills over first in the receiving flask?**

- |                                                |                                                               |
|------------------------------------------------|---------------------------------------------------------------|
| (A) Water, because its specific heat is higher | (B) Acetone, because its boiling point is significantly lower |
| (C) Both evaporate at identical rates          | (D) Neither, they form an azeotrope that cannot separate      |

**Q.11 What happens to the solubility of most solid solutes (like copper sulphate or potassium nitrate) in water as temperature increases?**

- |                                          |                                             |
|------------------------------------------|---------------------------------------------|
| (A) Solubility decreases linearly        | (B) Solubility increases significantly      |
| (C) Solubility immediately drops to zero | (D) Temperature has no effect on solubility |

**Q.12 When liquid air is warmed up slowly in a fractional distillation column, in what order do the gases distill out based on boiling points: Nitrogen (-196°C), Argon (-186°C), Oxygen (-183°C)?**

- |                                             |                                                          |
|---------------------------------------------|----------------------------------------------------------|
| (A) Oxygen first, then Argon, then Nitrogen | (B) Nitrogen first, followed by Argon, and lastly Oxygen |
| (C) Argon first, then Oxygen, then Nitrogen | (D) All three boil simultaneously                        |

**Q.13 A suspension of fine clay in water appears cloudy. When filtered through whatman filter paper:**

- |                                                 |                                                                                                |
|-------------------------------------------------|------------------------------------------------------------------------------------------------|
| (A) Both clay and water pass through completely | (B) Clay particles are retained on the filter paper while clear water passes into the filtrate |
| (C) The filter paper dissolves                  | (D) Clay turns into common salt                                                                |

**Q.14 An alloy of 70% copper and 30% zinc is called:**

- |            |           |
|------------|-----------|
| (A) Bronze | (B) Brass |
| (C) Solder | (D) Steel |

**Q.15 A solution contains 50 mL of pure alcohol dissolved in 150 mL of water. What is the volume percentage concentration of alcohol?**

- |           |           |
|-----------|-----------|
| (A) 33.3% | (B) 25.0% |
| (C) 20.0% | (D) 50.0% |

**SECTION B: ASSERTION - REASON QUESTIONS (Q.16 TO Q.20)****[5 MARKS]**

**Directions for Q16 to Q20:** Select the correct option from the choices given below:

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).  
 (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).  
 (C) Assertion (A) is true but Reason (R) is false.  
 (D) Assertion (A) is false but Reason (R) is true (or both are false).

**Q.16 Assertion (A):** Milk is a colloidal system and exhibits the Tyndall effect.

**Reason (R):** Colloidal fat and protein globules in milk have diameters comparable to visible light wavelengths, effectively scattering light rays.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).  
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).  
 (C) (A) is true but (R) is false.  
 (D) (A) is false but (R) is true.

**Q.17 Assertion (A):** Boiling point of an impure liquid is higher than that of the pure liquid.

**Reason (R):** Dissolved non-volatile solute particles lower the vapor pressure of the solvent, requiring a higher temperature to equal atmospheric pressure.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).  
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).  
 (C) (A) is true but (R) is false.  
 (D) (A) is false but (R) is true.

**Q.18 Assertion (A):** A separating funnel is used to separate a mixture of ethanol and water.

**Reason (R):** Ethanol and water are completely miscible liquids in all proportions.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).  
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).  
 (C) (A) is true but (R) is false.  
 (D) (A) is false but (R) is true.

**Q.19 Assertion (A):** Burning of a candle is considered both a physical and a chemical change.

**Reason (R):** Melting and vaporization of candle wax is physical, while combustion of wax vapor into  $\text{CO}_2$  and  $\text{H}_2\text{O}$  is chemical.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).  
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).  
 (C) (A) is true but (R) is false.  
 (D) (A) is false but (R) is true.

**Q.20 Assertion (A):** Centrifugation is used in diagnostic laboratories for blood and urine tests.

**Reason (R):** When spun rapidly, heavier blood cells settle to the bottom of the tube while lighter plasma remains on top.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).  
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).  
 (C) (A) is true but (R) is false.  
 (D) (A) is false but (R) is true.

**SECTION C: EXPERIMENTAL DATA & CASE STUDY (Q.21 TO Q.25)****[5 MARKS]****READING PASSAGE & EXPERIMENTAL SETUP:**

Case Study: Industrial Cryogenic Separation of Air

To produce pure liquid oxygen for medical cylinders, atmospheric air is passed through high-efficiency dust filters. It is then compressed under high pressure (200 atmospheres) and cooled by refrigeration pipes to remove water vapor and carbon dioxide (which would freeze and clog the machinery). The cold compressed air is expanded rapidly through a jet nozzle, causing Joule-Thomson cooling to  $-200^\circ\text{C}$  where air liquefies into liquid air. The liquid air enters a tall fractional distillation column.

**Q.21 Why must carbon dioxide and moisture be removed from the air before cooling below  $-78^{\circ}\text{C}$ ?**

- (A) They catch fire (B) They freeze into dry ice and ice crystals, blocking pipes and valves  
(C) They turn into toxic ozone (D) They evaporate instantly

**Q.22 What physical property difference allows the separation of nitrogen, argon, and oxygen in the column?**

- (A) Their chemical reactivity with copper (B) Differences in their boiling points  
(C) Differences in their magnetic attraction (D) Their rates of combustion

**Q.23 Which gaseous component collects at the very bottom of the distillation column as a liquid?**

- (A) Nitrogen (B) Oxygen  
(C) Argon (D) Helium

**Q.24 Is the production of liquid oxygen from atmospheric air a physical or chemical process?**

- (A) Exclusively chemical process (B) Physical process involving phase changes without altering molecular identities  
(C) Nuclear fusion reaction (D) Biological fermentation

**Q.25 Which gas comprises approximately 0.9% of dry air and boils at  $-186^{\circ}\text{C}$  between nitrogen and oxygen?**

- (A) Hydrogen (B) Carbon monoxide  
(C) Argon (D) Methane

**CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET B)**

|                    |                    |                    |                    |                    |
|--------------------|--------------------|--------------------|--------------------|--------------------|
| 01 (A) (B) (C) (D) | 02 (A) (B) (C) (D) | 03 (A) (B) (C) (D) | 04 (A) (B) (C) (D) | 05 (A) (B) (C) (D) |
| 06 (A) (B) (C) (D) | 07 (A) (B) (C) (D) | 08 (A) (B) (C) (D) | 09 (A) (B) (C) (D) | 10 (A) (B) (C) (D) |
| 11 (A) (B) (C) (D) | 12 (A) (B) (C) (D) | 13 (A) (B) (C) (D) | 14 (A) (B) (C) (D) | 15 (A) (B) (C) (D) |
| 16 (A) (B) (C) (D) | 17 (A) (B) (C) (D) | 18 (A) (B) (C) (D) | 19 (A) (B) (C) (D) | 20 (A) (B) (C) (D) |
| 21 (A) (B) (C) (D) | 22 (A) (B) (C) (D) | 23 (A) (B) (C) (D) | 24 (A) (B) (C) (D) | 25 (A) (B) (C) (D) |

Total Questions Attempted: \_\_\_\_ / 25

Marks Obtained: \_\_\_\_ / 25

Evaluator's Signature: \_\_\_\_\_

### OFFICIAL ANSWER KEY & IN-DEPTH EXPLANATIONS (SET B)

| Q. No. | Correct | In-Depth Conceptual & Numerical Solution                                                                                                                                                            |
|--------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q.1    | (B)     | For 100 g water, crystal mass $= 106 - 32 = 74$ g. For 50 g water, mass precipitated $= 74 / 2 = 37$ g.                                                                                             |
| Q.2    | (B)     | Let mass of salt be $x$ g. $\frac{x}{x + 180} = 0.10 \Rightarrow x = 0.10x + 18 \Rightarrow 0.90x = 18 \Rightarrow x = 20$ g.                                                                       |
| Q.3    | (B)     | $R_f = \frac{\text{Distance travelled by solute}}{\text{Distance travelled by solvent}} = \frac{4.8}{8.0} = 0.60$ .                                                                                 |
| Q.4    | (B)     | 1. Sublimation vaporizes $\text{NH}_4\text{Cl}$ leaving sand + salt. 2. Adding water dissolves salt; filtration removes insoluble sand. 3. Evaporation/crystallization recovers solid NaCl.         |
| Q.5    | (B)     | Glass beads supply a huge cool surface area where rising vapors repeatedly condense and re-evaporate, creating a sharp temperature gradient and enriching the vapor in the more volatile component. |
| Q.6    | (C)     | A gel is a colloid where a liquid is trapped in a solid matrix (dispersed phase: liquid, dispersion medium: solid, e.g. cheese, butter, jelly).                                                     |
| Q.7    | (B)     | Thermal Brownian motion keeps tiny colloidal particles suspended, while mutual electrostatic repulsion of like surface charges prevents agglomeration.                                              |
| Q.8    | (B)     | Rusting converts iron into hydrated iron(III) oxide ( $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ ), forming new chemical bonds through irreversible oxidation.                               |
| Q.9    | (B)     | Evaporating to dryness bakes impurities onto the crystals and can decompose heat-sensitive salts. Crystallization leaves impurities dissolved in the mother liquor.                                 |
| Q.10   | (B)     | Acetone has higher vapor pressure and lower boiling point ( $56^\circ\text{C}$ ), boiling first and condensing pure into the receiver while water remains in the flask.                             |
| Q.11   | (B)     | Dissolution of most solid salts is endothermic (absorbs heat); according to Le Chatelier's principle, higher temperature drives higher solubility.                                                  |
| Q.12   | (B)     | Nitrogen has the lowest boiling point ( $-196^\circ\text{C}$ ) and boils first, followed by Argon ( $-186^\circ\text{C}$ ), leaving liquid Oxygen ( $-183^\circ\text{C}$ ) at the base.             |
| Q.13   | (B)     | Suspension particles ( $>100$ nm) are larger than filter paper pores, so they are trapped as residue on the paper.                                                                                  |
| Q.14   | (B)     | Brass is a homogeneous solid solution alloy consisting of approximately 70% copper and 30% zinc. (Bronze is copper and tin).                                                                        |
| Q.15   | (B)     | $\text{Total volume} = 50 + 150 = 200$ mL. $\text{Volume \%} = \frac{50}{200} \times 100 = 25.0\%$ .                                                                                                |
| Q.16   | (A)     | Because dispersed fat and casein particles are 1-100 nm, they scatter incident photons, producing a bright visible beam path (Tyndall effect).                                                      |
| Q.17   | (A)     | Elevation of boiling point is a colligative property: non-volatile solute molecules occupy surface sites, lowering vapor pressure and raising boiling point.                                        |
| Q.18   | (D)     | Assertion is false: Separating funnels only work for *immiscible* liquids (like oil and water). Reason is true (ethanol and water are completely miscible).                                         |
| Q.19   | (A)     | Wax melts (solid to liquid) and vaporizes physically, while flame combustion chemically oxidizes hydrocarbon wax into carbon dioxide, water vapor, and soot.                                        |
| Q.20   | (A)     | Centrifugal acceleration forces denser particles (erythrocytes, leukocytes) outward and down, separating them cleanly from blood plasma.                                                            |
| Q.21   | (B)     | $\text{CO}_2$ sublimates at $-78.5^\circ\text{C}$ and water freezes at $0^\circ\text{C}$ ; solid ice and dry ice would choke cryogenic pipes.                                                       |
| Q.22   | (B)     | Fractional distillation separates liquefied gases based on boiling point differences ( $-196^\circ\text{C}$ , $-186^\circ\text{C}$ , $-183^\circ\text{C}$ ).                                        |

| Q. No. | Correct | In-Depth Conceptual & Numerical Solution                                                                                                 |
|--------|---------|------------------------------------------------------------------------------------------------------------------------------------------|
| Q.23   | (B)     | Oxygen has the highest boiling point (-183°C) among the three, so it remains liquid at the bottom while nitrogen vaporizes up.           |
| Q.24   | (B)     | Condensation, expansion, and fractional distillation are physical phase transitions; $O_2$ and $N_2$ molecules remain chemically intact. |
| Q.25   | (C)     | Argon is a noble gas constituting $\approx 0.93\%$ of atmospheric air with a boiling point of -186°C.                                    |